



Introduction to Biostatistics

MSBD520

Fall 2026

Nazirah Mohd Khairi, PhD

About Me

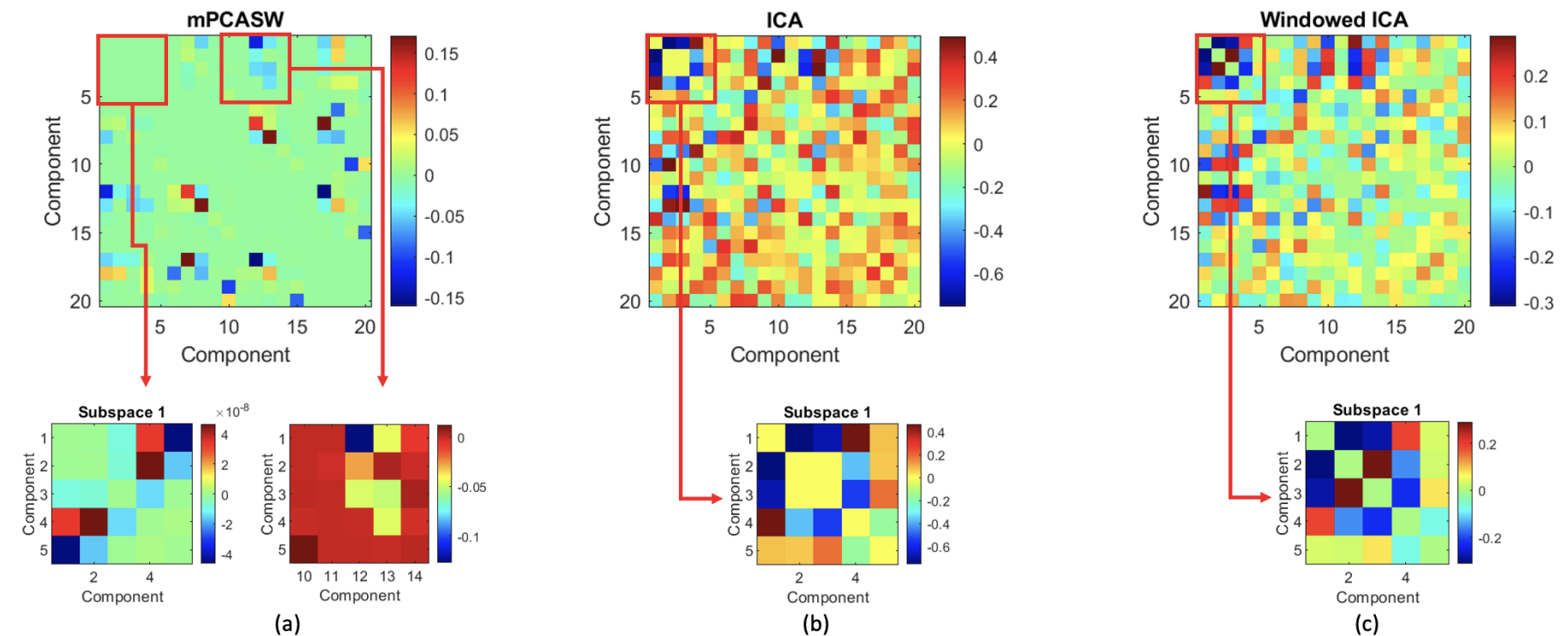
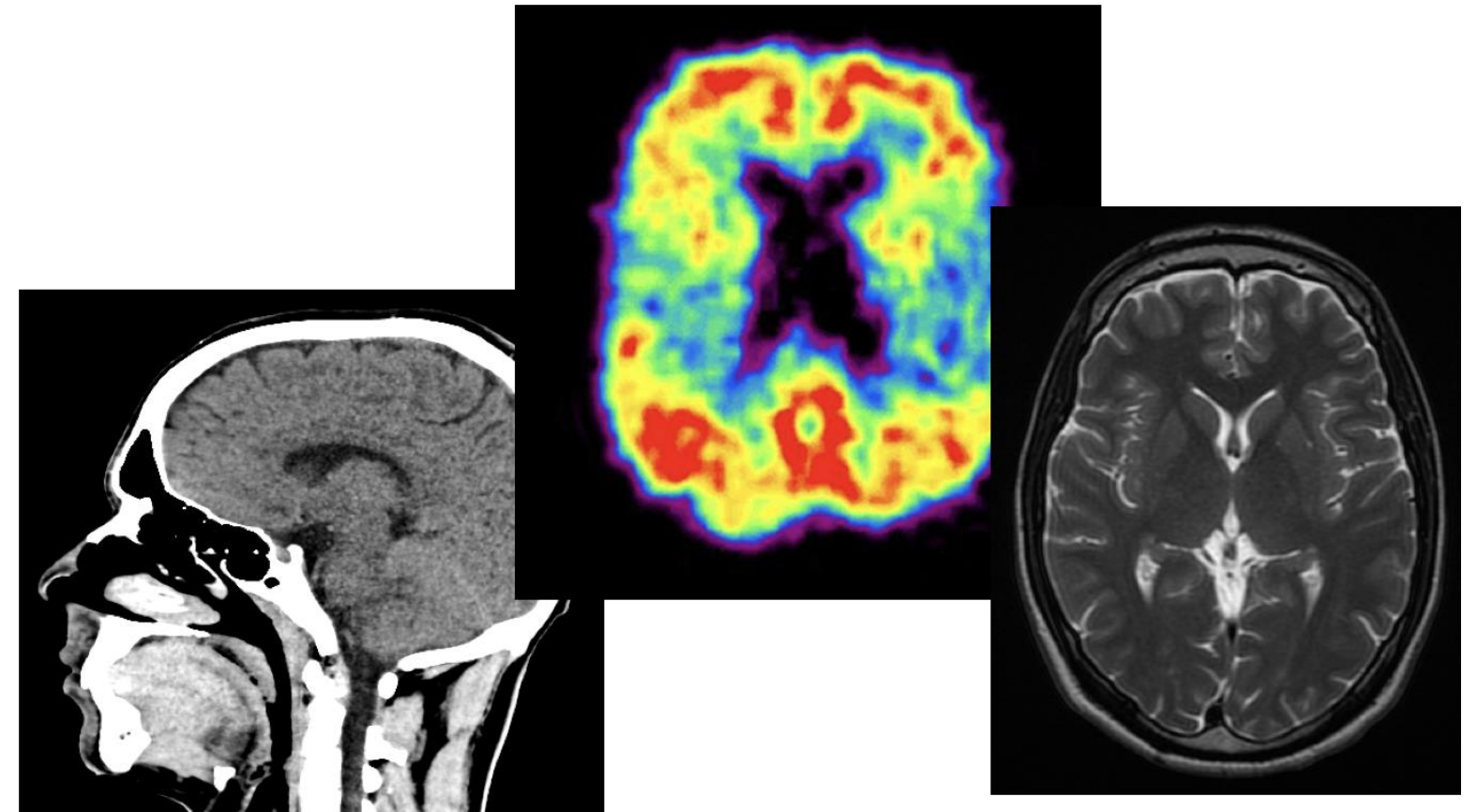
- Nazirah Mohd Khairi
- I am from Malaysia
- I have been in Nashville > 10 years (>15 in total)
- I love nature!
- I use MATLAB > 10 years
- I can also speak Python, Bash, Octave (used VB, C++, Perl a while ago).



About Me

- Office Hours: TBD. I will update the Blackboard and send announcement after confirmed.
- Office location: 3401 West End Ave, 3rd floor (email me)
- Email address:
nazirah.mohdkhairi@mmc.edu

Note: emailing me directly will be the best way to reach me.



Warming up...

Let's get to know each other 😊

Your
background

What are you doing right now
other than taking this
program?

Current city

Research/career interest

Fun facts about you or
anything interesting
during summer

Course Information

- Course Title: Introduction to Biostatistics
- Class Time: 5:30pm – 8:30pm (CT); Thursdays
- Class Location: MSBD 520 Blackboard Class > Launch Class [Chrome or Edge is best to use].

Prerequisites

- Elementary Statistics, Linear Algebra, Calculus, Discrete Math.
- Have R/Python/SAS and its environment installed in your computer – we will use this for assignments and project development.
- Getting familiar with R/Python/SAS

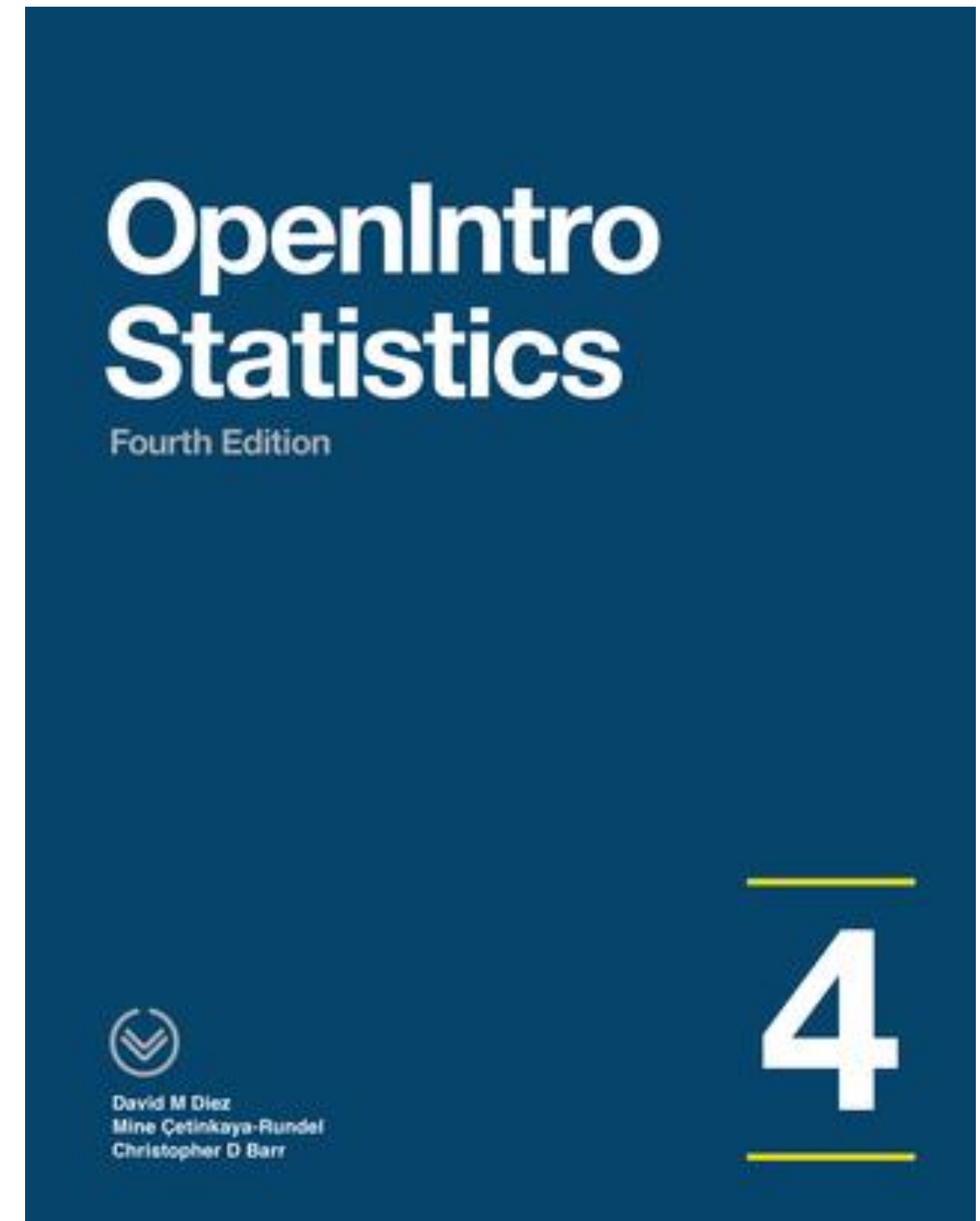
Course Objective

This course is designed to meet the following objectives:

- Describe the **basic concepts** of probability, random variation and commonly used statistical probability distributions.
- **Apply descriptive and inferential methodologies** according to the type of study design for answering particular research questions.
- Recognize and implement parametric and nonparametric analysis methods for **testing hypotheses**.
- **Build and interpret mathematical models** of real-world phenomena in and across multiple disciplines, including linear algebra, discrete mathematics, probability, and calculus, with an emphasis on applications in data science and data engineering.
- Understand and utilize **analysis of variance designs** with the use of contrasts and multiple comparisons procedures.
- Utilize variable selection methods, modeling techniques, regression diagnostics, and model validation.

Books

- **(REQUIRED)** OpenIntro Statistics; David Diez, Mine Cetinkaya-Rundel, Christopher Barr; 4th Edition
- Introductory Statistics for the Life and Biomedical Sciences, First Edition by Julie Vu and David Harrington
- Fundamentals of BIOSTATISTICS: 7th Edition by Bernard Rosner
- Think Stats by Allen B. Downey, Second Edition (with Python code)



Assignment Details

- **Pre/In-class activities:** Work done individually, in pairs. We might use some pre-class work in our in-class activities. We will try to learn stats through hands-on examples 😊
- **Homework:** exercises similar to the in-class examples but more extensive with interpretation questions to enhance understanding.
- **Quiz:** "Small exams" to cover recent topics. Quizzes will be done through Blackboard with multiple-choice and short-answer questions.
- **Project:** Bigger task than the homework.
- **Exam:** One final exam due on the last day of the semester.

Grading Outline

Assessment	Weight percentage
Attendance	5%
Homework	20%
Pre/In-class activities	15%
Quiz	20%
Project	20%
Exam	20%
TOTAL	<u>100</u>

Attendance and Submissions

- **Attendance:**
 - Attendance is required for all sessions. More details are in the syllabus.
 - If you cannot attend:
 - And you know ahead of time – you are required to let me know as early as possible, not on the day of the class.
 - If it is an emergency – please let me know as soon as you can. Send me applicable documents (doctor's note, report, etc.) as well.
- **Submissions:**
 - You are responsible for meeting all the deadlines.
 - If you happen to miss classes, you need to check the recording, the slides, or assignment information and deadlines on Blackboard.
 - If you need extra time due to an emergency, please contact me ASAP. Otherwise, late penalty will be enforced.

Programming Platform

We will mostly use Python as the course programming language. But you are welcome to use other languages you prefer (just let me know upfront).

.py



For easy reference, we will use Python notebooks *ipynb* (specifically Jupyter notebooks) for notes and other class work.

- There are many ways to open and run the notebook.
- I will show Google Colab (no installation required), Jupyter, and VS Code as examples.



Programming Platform

How to check if you have Python on your computer?

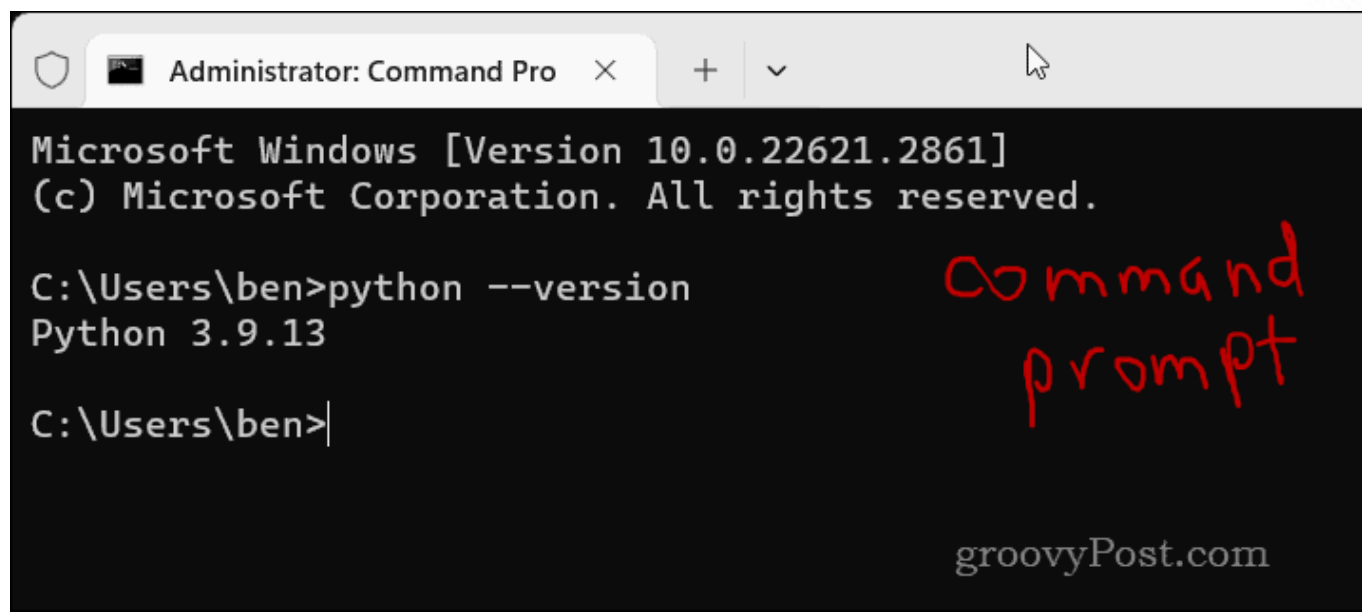
MacOS: Use Terminal, and type **python3 --version**

Windows OS: Use the command prompt or Windows PowerShell, and type **python3 --version**



If you get an error, we will need to install Python.

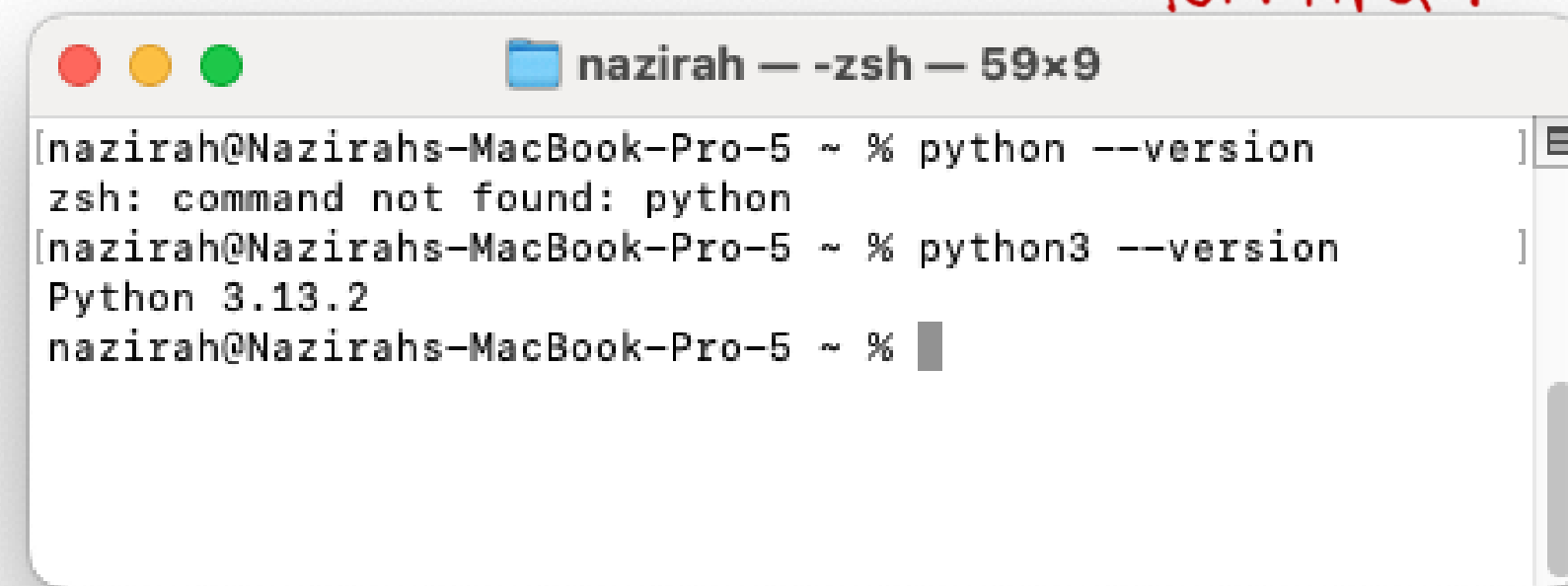
P.S.: You can use Google Colab without Python installed since it is online.

A screenshot of a Windows Command Prompt window titled "Administrator: Command Pro". The text inside shows the command prompt version (10.0.22621.2861) and the user running the command "python --version", which returns "Python 3.9.13". A red handwritten note "command prompt" is written over the output. The URL "groovyPost.com" is visible at the bottom right.

```
Administrator: Command Pro
Microsoft Windows [Version 10.0.22621.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ben>python --version
Python 3.9.13

C:\Users\ben>
```

A screenshot of a MacOS Terminal window titled "nazirah - zsh - 59x9". The text shows the user running "python --version" which results in "zsh: command not found: python", and then running "python3 --version" which returns "Python 3.13.2". A red handwritten note "terminal" is written above the window.

```
nazirah - zsh - 59x9
[nazirah@Nazirahs-MacBook-Pro-5 ~ % python --version
zsh: command not found: python
[nazirah@Nazirahs-MacBook-Pro-5 ~ % python3 --version
Python 3.13.2
nazirah@Nazirahs-MacBook-Pro-5 ~ %
```

Can I use AI in this course?

In short → YES! But with limited conditions.

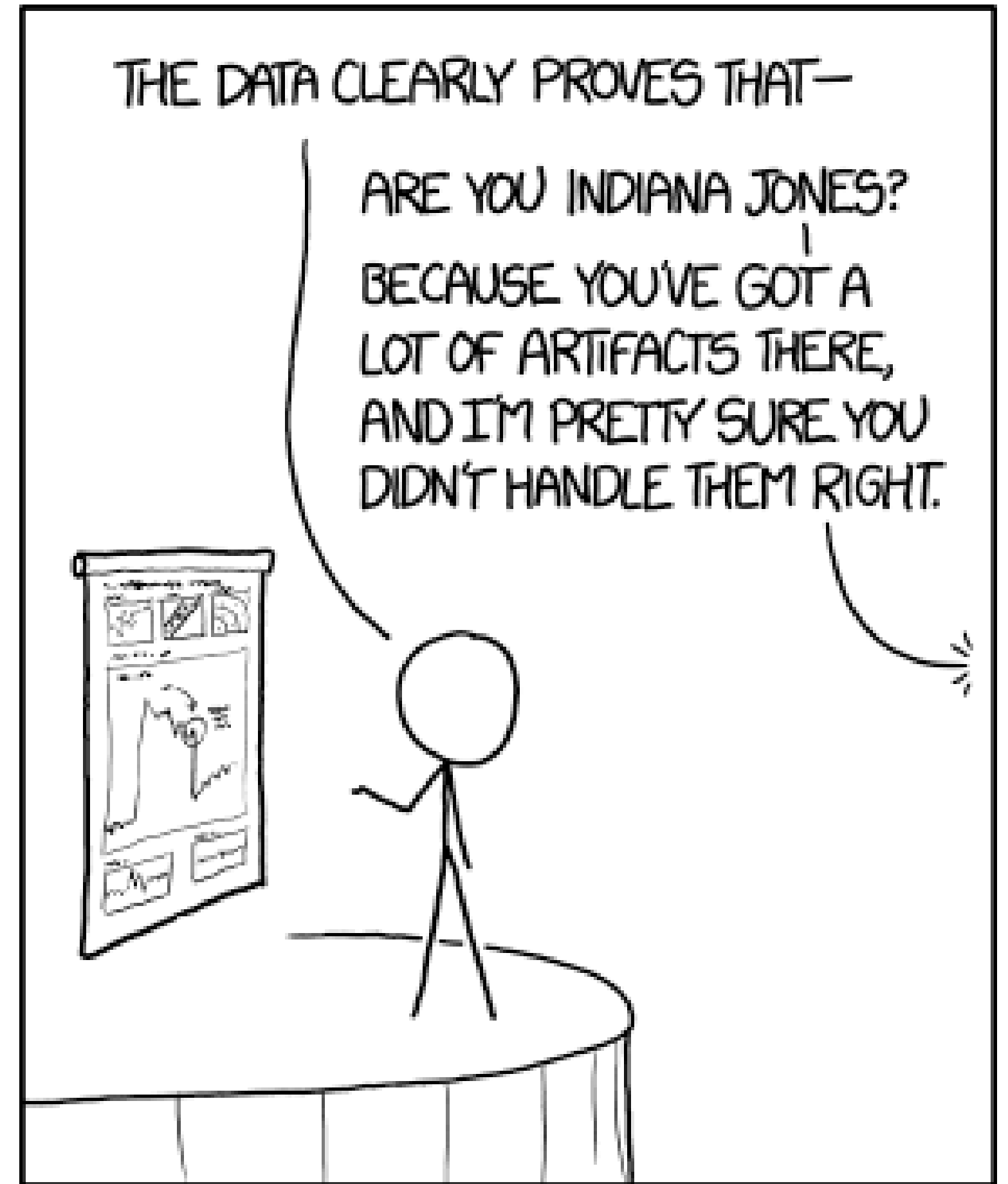
All ideas, interpretations, and scholarly content must be solely by the student. Students are fully responsible for the accuracy, integrity and originality of the submitted work.

Writing
Enhancement

Grammar Check
Tool

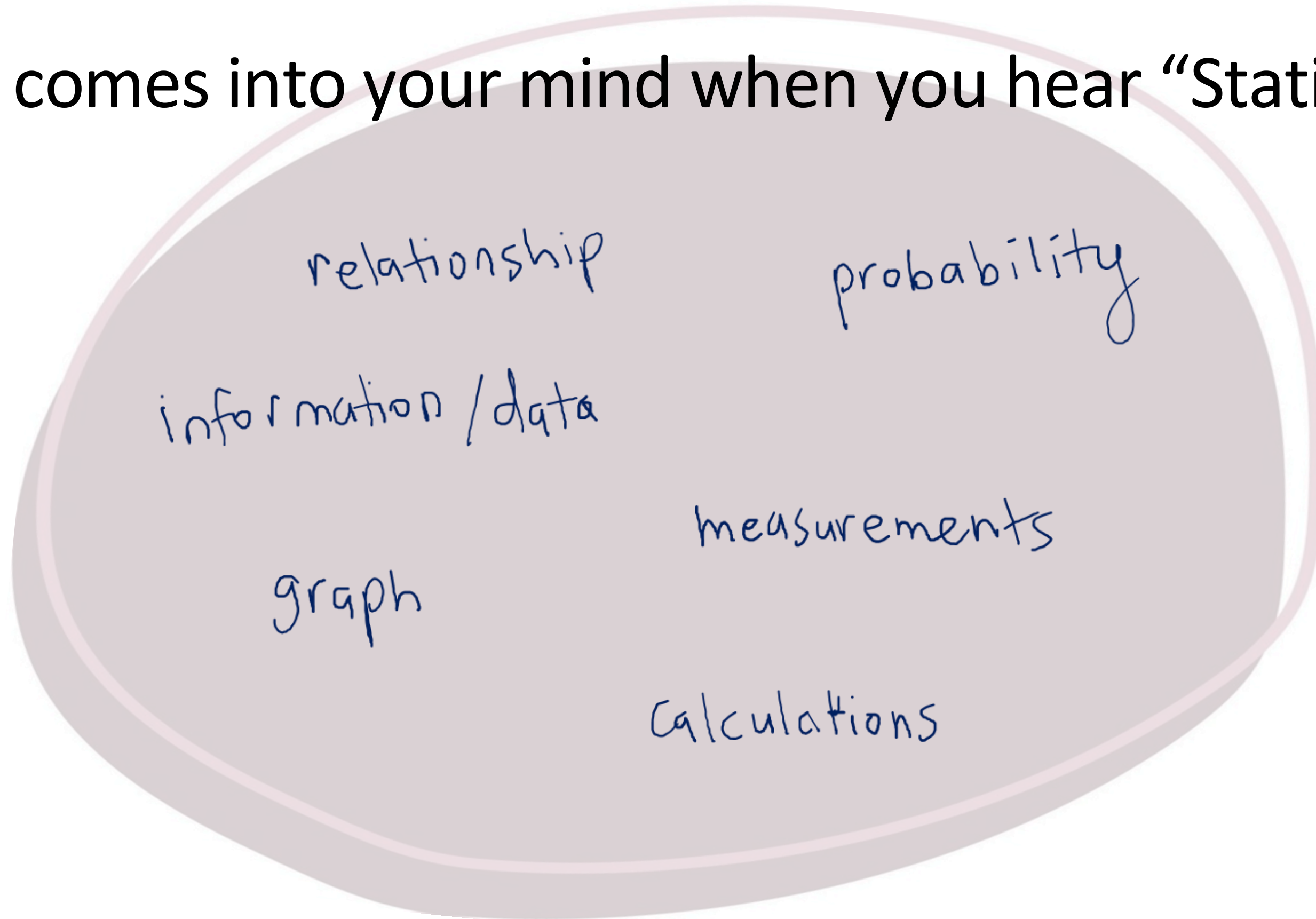
Key point: You are accountable for your work.

Do you think you
do statistics
right?



What is statistics?

What comes into your mind when you hear “Statistics”?



An Example

There are millions of airline passengers in the United States.

↙
population of interest

Question:

What is their average age?

How do you answer this question?



Statistics

Statistics is the science concerned with developing and studying methods for collecting, analyzing, interpreting, and presenting empirical data.

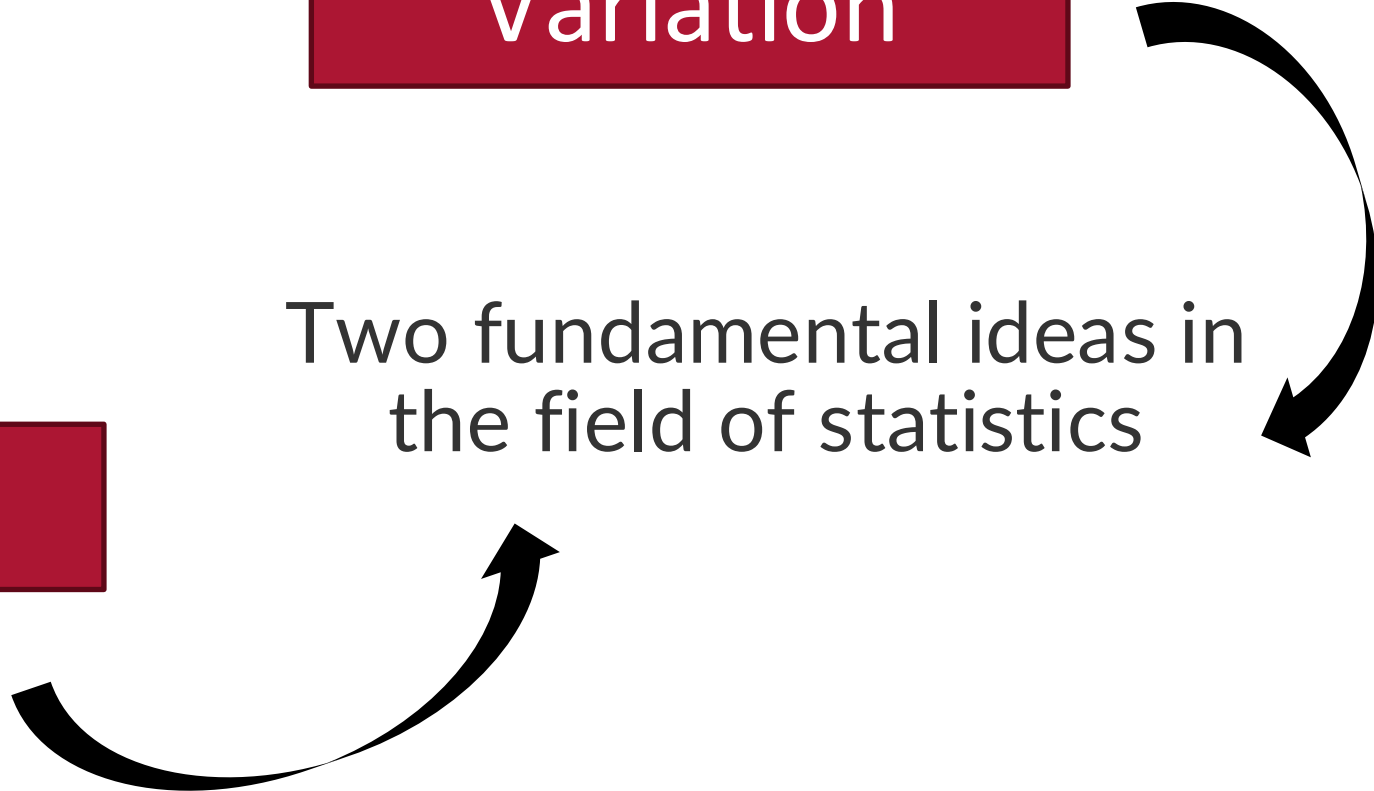
Probability

a mathematical language
used to discuss uncertain
events

Variation

Uncertainty

Two fundamental ideas in
the field of statistics

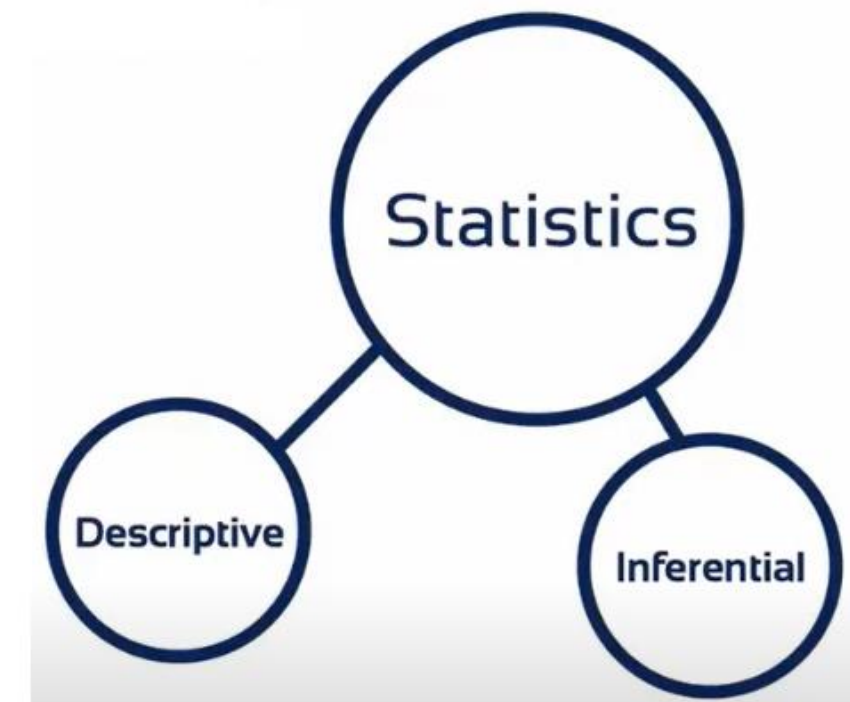


Statistics

- Biostatistics is the application of statistical principles in medicine, public health, biology, life science.

There are two main branches of statistics:

- **Descriptive statistics:** Summarizing and communicating information about a group of numbers (data). They refer only to the actual data available.
- **Inferential statistics:** Drawing conclusions based on the data collected, and making predictions that go beyond the immediate data.



How do we do the statistics?

What are the steps we have to take?

What are we going to look in this course?



Understanding of data



Summarizing data



Probability



Distributions



Introduction to inference

Before next week class session...

- Get yourself familiarized with –
 - Blackboard and Class App.
 - Syllabus - course policy
 - OpenIntro Statistics Book and website
 - Python/R/Programming platform for you to work on.
- Submit In-Class 1: Deadline (Aug 19)